

WEEK 10

TUTORIAL - DURATION MODELS, CENSORING, TRUNCATION, SELECTION

APPLIED STATISTICAL ANALYSIS II

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SPRING 2026

ROADMAP FOR TODAY

- Understand survival/duration data and censoring
 - ▶ Interpret key functions: survival, density, hazard
 - ▶ Kaplan–Meier and Cox proportional hazards in practice
 - ▶ Grasp survival–Poisson equivalence
- Build intuition for Tobit and Heckman selection models

SURVIVAL / DURATION SETUP

- **Outcome:** time-to-event with a clear start, time scale, and event
- **Right-censoring:** observation stops before event, only know $T > t$
- **Left-truncation:** entry after process starts $T \mid T > t_0$
- In R, survival outcomes are coded with `Surv()`

KEY FUNCTIONS: $S(t)$, $f(t)$, $\lambda(t)$

- Survival: $S(t) = \Pr(T > t)$, probability event has not occurred by t
- Density: $f(t)$ is the unconditional event rate, $f(t) = -S'(t)$
- Hazard:
$$\lambda(t) = \lim_{dt \rightarrow 0} \Pr\{T \in (t, t + dt) \mid T > t\} / dt = f(t) / S(t)$$
- Link: $S(t) = \exp\left(-\int_0^t \lambda(u) du\right)$; constant λ gives exponential survival

KAPLAN-MEIER CURVES

- Nonparametric estimate of $S(t)$; step function that drops at event times
- At each event time: drop by k/n , with k events among n at risk
- Censored observations reduce the at-risk set but do not create drops
- Best for descriptive comparisons (e.g., treatment vs. control), not regression with covariates

COX PROPORTIONAL HAZARDS

- Model: $h(t | X) = h_0(t) \exp(X\beta)$ with unspecified baseline hazard $h_0(t)$
- $\exp(\beta_j)$ is the hazard ratio for a one-unit increase in X_j , holding others fixed
- Example: male mortality in 1800s using
`coxph(Surv(enter,exit,event) ~ ses + birthdate)`

SURVIVAL-POISSON CONNECTION

- If hazard is constant over intervals, event counts follow $Y_i \sim \text{Poisson}(\lambda_i t_i)$
- Model: $\log E[Y_i] = \beta_0 + \beta_1 X_i + \log(t_i)$, with $\log(t_i)$ as offset (person-time)
- With suitably fine time-splitting, Poisson with offset reproduces Cox PH estimates
- Useful for directly modeling incidence rates and handling time-dependent covariates

TOBIT MODELS

- Latent outcome Y_i^* observed only above (or below) a censoring threshold; others are censored at the limit
- Same covariates and coefficients govern both $\Pr(\text{censored})$ and the latent outcome
- Coefficients describe effects on $E[Y_i^*]$; marginal effects on observed Y_i are scaled by $\Pr(\text{uncensored})$
- Ex: academic aptitude scores censored at 200 and 800, estimated via `tobit` or `survreg`

HECKMAN SELECTION MODELS

- Address selection bias when the outcome is observed only for a non-random subset
- Two equations: selection (participation) equation and outcome (e.g., wage) equation
- Selection equation estimated via Probit; inverse Mills ratio enters outcome equation
- Ex: Mroz (1987) women's wages and labor-force participation using `heckit`

BIG PICTURE

■ Duration models:

- ▶ Structure time-to-event data and censoring via $S(t)$, $f(t)$, and $\lambda(t)$
- ▶ Cox PH and Poisson with offsets offer flexible ways to model hazards and rates

■ Tobit and Heckman models:

- ▶ Extend linear models to censoring and selection
- ▶ Make estimates more precise + causal interpretation more credible